

DLS Buddy — Theory and Equations Guide

Applies to v1.0.0 and later

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Equations are numbered (1), (2), ... and rendered in LaTeX. Whenever a new analysis is added to the program, its equation(s) are added here.

1. Physics and Calibration

All physical formulas live in `physics/constants.py`; the analysis code never re-implements them.

1.1 Scattering Vector

$$q = \frac{4\pi n}{\lambda} \sin\left(\frac{\theta}{2}\right) \quad (1)$$

where λ is the vacuum wavelength of the laser, n is the solvent refractive index at that wavelength and temperature, and θ is the scattering angle. Both λ and n are always user-supplied. q is computed in nm^{-1} for SLS and in m^{-1} for the Stokes–Einstein step. This is the standard definition for solution scattering [1].

1.2 Stokes–Einstein Relation

The hydrodynamic radius follows from the translational diffusion coefficient $D = \Gamma/q^2$ (Γ is the field-autocorrelation decay rate) via the **Stokes–Einstein relation** [1]:

$$R_h = \frac{k_B T}{6\pi\eta D} \quad (2)$$

where k_B is the Boltzmann constant, T the absolute temperature, and η the solvent dynamic viscosity. T and η are always user-supplied. A single-angle measurement gives the **apparent** R_h ; extrapolation to $q \rightarrow 0$ and $c \rightarrow 0$ is handled in the analysis module.

1.3 Refractive-Index Correction

When the sample solvent differs from the calibration standard, the excess Rayleigh ratio is corrected by

$$f = \left(\frac{n_{\text{solvent}}}{n_{\text{standard}}} \right)^2 \quad (3)$$

applied as $\Delta R\theta = R\theta_{\text{measured}} \cdot f$. When solvent = standard, $f = 1$ exactly. This correction is **mandatory** whenever the solvent is not the calibration liquid.

1.4 Optical Constant

The prefactor in the Zimm/Debye/Berry equations (vertically polarized geometry):

$$K = \frac{4\pi^2 n^2 (dn/dc)^2}{N_A \lambda^4} \quad (4)$$

with NA the Avogadro number. Units are $\text{mol}\cdot\text{mL}\cdot\text{g}^{-2}\cdot\text{cm}^{-1}$ (λ converted $\text{nm}\rightarrow\text{cm}$ internally), consistent with c in g/mL and $R\theta$ in cm^{-1} .

Why dn/dc matters. $K \propto (dn/dc)^2$, so the refractive-index increment enters the molecular weight **squared**. For a low-contrast system a small dn/dc makes the scattering weak and amplifies its error. It is user-supplied and must never be guessed.

1.5 Toluene Rayleigh Ratio (geometry-aware)

All SLS calibration uses the temperature- and geometry-corrected toluene Rayleigh ratio — **never** the value stored in instrument software. The base quantity is the polarized ratio R_{VV} , temperature-corrected and then converted to the measurement geometry:

$$R_{VV}(T) = R_{VV}(25^\circ\text{C}) [1 + \alpha (T - 25)], \quad \alpha = +0.0043 \text{ }^\circ\text{C}^{-1} \quad (5)$$

$$R_{VU} = R_{VV} (1 + \rho_v), \quad R_{VH} = R_{VV} \rho_v \quad (6)$$

where $\rho_v = IVH/IVV$ is the depolarization ratio, $\rho_v(T) = 0.346 - 0.00135 (T - 25)$ unless overridden.

Wavelength	R_{VV} at 25 °C (cm^{-1})	Source
532 nm	2.34×10^{-5}	[2]
660 nm	8.456×10^{-6}	[3], Table 2

Geometry must match the calibrant. The temperature coefficient α is **positive** — the Rayleigh ratio *rises* with temperature. The geometry (VV / VU / VH) must match how the calibrant was measured, or the absolute scale is wrong by $\approx (1 + \rho_v) \approx 1.35$. An instrument with a vertical laser and **no analyzer** is **VU** geometry. Example: $R_{VU}(532 \text{ nm}, 25^\circ\text{C}) = 2.34 \times 10^{-5} \times (1 + 0.346) \approx 3.15 \times 10^{-5} \text{ cm}^{-1}$.

Other wavelengths. Add a properly sourced R_{VV} to `_TOLUENE_RVV_25C` in `physics/constants.py`. Do **not** interpolate — the wavelength dependence is steep: toluene’s Rayleigh ratio scales as $\approx \lambda^{-4.17}$, steeper than the λ^{-4} Rayleigh law and reducing to $\approx \lambda^{-4}$ only once the $n(\lambda)$ dispersion is separated out [4].

2. Data-Quality and Cross-Analysis Theory

2.1 Intensity-trace diagnostics

Two quantities reported by the trace diagnostics (the procedure is in the Quickstart) rest on the shot-noise model of photon counting.

Outlier band. Points outside the shot-noise envelope are flagged:

$$\text{flag if } |I - \bar{I}| > k\sqrt{\bar{I}}, \quad k = 1 \text{ (default)} \quad (7)$$

For a Poisson (shot-noise-limited) process the standard deviation equals $\sqrt{\text{mean}}$, so this is the expected $\pm k\sigma$ band; the flagged fraction is a compact quality metric. Note $\sqrt{\text{mean}}$ equals the true shot-noise σ only for raw photon counts; for count rates (cps) the envelope is a useful heuristic.

Fano factor. The count-rate histogram is fitted with a Gaussian and/or Poisson distribution; the key diagnostic is the Fano factor (variance/mean in count space):

$$F = \frac{\sigma^2}{\mu} \quad \begin{cases} \approx 1 & \text{shot-noise limited (ideal Poisson)} \\ > 1 & \text{excess fluctuations (slow modes, dust, drift)} \end{cases} \quad (8)$$

2.2 $\rho = R_g/R_h$

The shape parameter links the radius of gyration (R_g , from SLS) and the hydrodynamic radius (R_h , from DLS) of the same sample:

$$\rho = \frac{R_g}{R_h} \quad (9)$$

Both should be infinite-dilution values for ρ to be meaningful; the program checks that the two measurements share polymer, solvent, and temperature. The architecture reference values below are well-known guides, not sharp boundaries [1].

ρ	Architecture
≈ 0.775	Hard sphere (compact, near-spherical)
≈ 1.0	Hyperbranched polymer, microgel, or vesicle
$\approx 1.5\text{--}1.8$	Linear random coil (θ -solvent ≈ 1.5 , good solvent ≈ 1.78)
> 2	Extended/rigid-rod, or aggregated/polydisperse

These are guides, not hard boundaries. ρ is **flagged apparent** if *either* input is **apparent** — a single-condition R_g or R_h is never silently combined with a thermodynamic one. For a physically meaningful shape parameter, prefer the thermodynamic (infinite-dilution) source on both axes.

2.3 Scaling laws (R_g – M_w , A_2 – M_w)

Across a homologous series (one polymer/solvent, varying molecular weight), size and the second virial coefficient follow power laws in M_w :

$$R_g = k_g M_w^\nu \quad (10)$$

$$A_2 = k_a M_w^{-a} \quad (11)$$

The program fits each as a straight line in log–log space; the **exponent** is the physically meaningful quantity and is independent of the (calibration-dependent) prefactor. Typical flexible-coil values: $\nu \approx 0.588$ in a good solvent, $1/2$ at the θ -point, $1/3$ for a collapsed globule or hard sphere; $a \approx$

0.2–0.3 [5], [6]. The fit needs at least two samples with the relevant pair. Below each plot the tab prints a **plain-language reading** of the fitted exponent (e.g. “ $\nu \approx 3/5 \rightarrow$ swollen coil, good solvent”), mirroring the ρ table’s architecture hint — a guide, not a hard boundary.

3. Dynamic Light Scattering (DLS) Analysis

The DLS module extracts particle dynamics — decay rates, diffusion coefficients, and hydrodynamic radii — from a measured $g_2(\tau)-1$. All methods fit $g_2(\tau)-1$ **directly** (β floating) and carry the Siegert factor of 2 explicitly, so a reported decay rate Γ is the physical g_1 rate; then $D = \Gamma/q^2$ and Rh follows from Stokes–Einstein, Eq. (2) (which needs the viscosity; decay rates and diffusion coefficients do not).

The two correlation-to-quantity conversions used below appear first as data forms when loading a generic file. For the **g2** form, the baseline is removed by

$$g_2(\tau) - 1 = \frac{G(\tau) - B}{B} \quad (12)$$

For the **g1** form, the **Siegert relation** [1] converts the field autocorrelation to the measured quantity:

$$g_2(\tau) - 1 = \beta |g_1(\tau)|^2 \quad (13)$$

where β is the instrument coherence factor (an optical property, typically 0.01–1.0 for standard goniometers), determined independently.

3.1 Parametric Methods

Cumulant expansion [7]. The standard first analysis. Taking the log of the baseline-subtracted signal and fitting a polynomial in τ :

$$\ln[g_2(\tau) - 1] = \ln \beta - 2\bar{\Gamma} \tau + \mu_2 \tau^2 - \frac{1}{3} \mu_3 \tau^3 + \dots \quad (14)$$

The mean decay rate $\bar{\Gamma}$ gives $\bar{D} = \bar{\Gamma}/q^2$ and Rh ; the second cumulant gives the **polydispersity index**

$$\text{PDI} = \frac{\mu_2}{\bar{\Gamma}^2} \quad (15)$$

and the third (order-3 fit) the skewness. The cumulant result is reliable only up to $\text{PDI} \approx 0.3$; the program flags whether this is satisfied. Above it, use a distribution method. Equation (14) is the **linear** (log-polynomial) method; it assumes the baseline is exactly zero and is fit by weighted linear least squares.

Nonlinear cumulant [8]. The program’s **default** cumulant method fits g_2-1 *directly* by nonlinear least squares, expanding about the mean rate (Friskens’s moments-about-the-mean form) with a **floating baseline** B :

$$g_2(\tau) - 1 = B + \beta \left[e^{-\bar{\Gamma}\tau} \left(1 + \frac{1}{2}\mu_2\tau^2 - \frac{1}{6}\mu_3\tau^3 + \dots \right) \right]^2 \quad (16)$$

It yields the same $\bar{\Gamma}$, μ_2 , PDI and Rh as the linear method but is more robust to baseline drift and noisy/low-count data, because B and the coherence β are fit rather than assumed and the whole windowed correlogram is used (no amplitude cutoff). Expanding about $\bar{\Gamma}$ (rather than about $\tau = 0$, as the log-polynomial does) keeps it stable at large τ . The baseline is constrained to small drift about the long-time tail so the fit cannot absorb the decay into B ; if it fails to converge it falls back to the linear fit, flagged. The method is a **global setting** (**Settings** $\rightarrow$ **Cumulant method**, default *Nonlinear*) that applies to every cumulant-based step (per-measurement cumulant, Γ -vs- q^2 , replicate averaging); changing it clears existing cumulant-based results so the workspace stays self-consistent. **Uncertainty:** like the linear method, the nonlinear fit reports **no $\pm$ from a single correlogram** (its lag channels are correlated — Schätzel 1990 — so the fit covariance under-reports); the honest DLS $\pm$ comes from replicate averaging (Eq. 58), per invariant 8.

Analysis window: delay bounds + leading-channel skip. Every fit (cumulant, exponentials, KWW, NNLS/CONTIN/lognormal) selects which correlogram points it uses through one shared window with two complementary restrictions, applied in order:

1. a **leading-channel skip** — drop the first N correlator channels *by index* (the shortest, finest-spaced lags). The first few channels of a multi- τ correlator carry detector **afterpulsing** and dead-time artefacts that are not diffusion: a spuriously high first channel mimics a fast decay and biases the fit toward a falsely small Rh . The skip is an integer set in **Settings** $\rightarrow$ **Skip initial lag channels** (default 0 = keep all);
2. a physical **delay-time window** $[\tau_{\min}, \tau_{\max}]$ in time.

The two **compose as an intersection** — the effective first fitted point is the later of “channel N ” and “first channel with $\tau \geq \tau_{\min}$ ”. The skip is applied **first**, so a corrupted leading channel cannot distort the amplitude cutoff/intercept (cumulant) or the β -to- $\tau \rightarrow 0$ extrapolation (distributions). Use it sparingly with distribution methods: the short lags constrain the *fast / small* end of the size distribution, so an aggressive skip that a cumulant *mean* merely nudges can erase a genuine small-particle population entirely.

Single exponential — a monodisperse sample (equivalent to the first cumulant for clean data):

$$g_2(\tau) - 1 = \beta e^{-2\Gamma\tau} \quad (17)$$

Double exponential — two discrete modes through the full Siegert relation (appropriate when a polymer and a slow aggregate coexist); the fit reports whether it converged:

$$g_2(\tau) - 1 = \beta \left[f_1 e^{-\Gamma_1\tau} + (1 - f_1) e^{-\Gamma_2\tau} \right]^2 \quad (18)$$

Kohlrausch–Williams–Watts (stretched exponential) — broadly distributed or glassy dynamics, stretch exponent $s \in (0, 1]$ ($s = 1$ recovers a single exponential):

$$g_2(\tau) - 1 = \beta \exp[-2(\tau/\tau_c)^s], \quad \langle \tau \rangle = \frac{\tau_c}{s} \Gamma\left(\frac{1}{s}\right) \quad (19)$$

Both the characteristic time τc and the mean time $\langle \tau \rangle$ (via the gamma function) are reported, each with its hydrodynamic radius.

3.2 Distribution Methods

These recover a full distribution of decay rates without assuming a functional form. The field autocorrelation is a weighted sum of exponentials,

$$g_2(\tau) - 1 = \beta \left[\int G(\Gamma) e^{-\Gamma \tau} d\Gamma \right]^2, \quad (20)$$

discretized on a grid of hydrodynamic radii (default 1–1000 nm, log-spaced). We invert the **field ACF** g_1 directly [9, Sec. 8.11]: the single- Γ kernel is $A[m, n] = \exp(-\Gamma n \tau m)$ and the data vector is g_1 recovered from the Siegert relation, $\mathbf{y} = \text{sign}(u) \sqrt{|u|}$ with $u = (g_2 - 1 - \text{baseline})/\beta$ — a **sign-preserving** square root, so long-delay noise stays zero-mean rather than rectifying into a positive pedestal that a non-negative fit would read as spurious large- R_h weight. The reconstruction back to measured space is $g_2 - 1 = \beta(\mathbf{Ax})^2 + \text{baseline}$. Both methods need β and a baseline (estimated automatically — β by extrapolating to zero delay, the baseline from the long-delay tail — and overridable; the values used are always reported).

The residual is **statistically weighted** by the delta-method inverse variance of the recovered g_1 : propagating a (locally uniform) $g_2 - 1$ noise through the square root gives $\text{Var}(g_1) \propto 1/|u|$, so the per-lag weight is $w \propto g_1^2$ ($\mathbf{M} = \text{diag}(w)$), down-weighting the noise-dominated long-delay channels. The overall scale cancels in the fit and in the α selectors, so the weight is **parameter-free**. It is a **fit-only precision map** — **no statistical $\pm$ is reported** from it: the distributions stay uncertainty-free (a single correlogram’s lag channels are correlated, Schätzel 1990, and the off-diagonal covariance is large, so an honest $\pm$ is deferred to replicate averaging — §6.5).

- **NNLS** solves $\min_{\mathbf{x} \geq 0} \|\mathbf{M}^{1/2}(\mathbf{Ax} - \mathbf{y})\|^2$ with $\mathbf{x} \geq 0$ and no smoothing. It resolves well-separated peaks but produces spiky, noise-sensitive distributions.
- **CONTIN** [10] adds Tikhonov regularization:

$$\min_{\mathbf{x} \geq 0} \|\mathbf{M}^{1/2}(\mathbf{Ax} - \mathbf{y})\|^2 + \alpha^2 \|\mathbf{Lx}\|^2 \quad (21)$$

with $\mathbf{L}$ the second-difference operator. The regularization strength α is chosen automatically by one of **three** selectors over the same sweep (the full sweep is returned and a fixed α can be supplied); each records on the result which selector chose it. CONTIN is generally better for broad distributions and real data.

GCV — *generalized cross-validation* (default; [11]). GCV estimates the leave-one-out prediction error without knowing the noise level. For our weighted Tikhonov fit, with $\text{tr}(\mathbf{I} - \mathbf{A}_\alpha) = N_y - \text{NDF}(\alpha)$,

$$\text{GCV}(\alpha) = \frac{N_y V(\alpha)}{(N_y - \text{NDF}(\alpha))^2}, \quad V(\alpha) = \|\mathbf{M}^{1/2}(\mathbf{Ax}_\alpha - \mathbf{y})\|^2, \quad (22)$$

where $V(\alpha)$ is the weighted residual, $\text{NDF}(\alpha)$ the Tikhonov effective degrees of freedom (Eq. (25)), and N_y the number of lag channels; the chosen α is the minimizer. GCV is σ^2 -free

and asymptotically optimal (Golub, Heath & Wahba 1979) and is the robust default for this ill-conditioned g_1 kernel — it resolves noisy multimodals the L-curve over-smooths. Its one documented failure mode is a **flat** minimum ([12], §7.4): a minimum pinned to the **high- α** end of the sweep signals over-regularization and raises a visible caution (widen the range, or inspect the L-curve); a low- α (floor) pick is legitimate when clean data needs little smoothing.

L-curve corner ([12], Eq. (7.24); corner-finding algorithm [13]). Plot the trade-off curve $(\log\|\mathbf{A}\mathbf{x}_\alpha - \mathbf{y}\|, \log\|\mathbf{L}\mathbf{x}_\alpha\|)$ and pick the α of **maximum curvature** — the corner of the “L”. A cubic spline is fitted through the discrete points and its analytic curvature maximised:

$$\kappa(\alpha) = \frac{\zeta'\eta'' - \zeta''\eta'}{((\zeta')^2 + (\eta')^2)^{3/2}}, \quad \zeta = \log\|\mathbf{A}\mathbf{x}_\alpha - \mathbf{y}\|, \quad \eta = \log\|\mathbf{L}\mathbf{x}_\alpha\|, \quad (23)$$

primes denoting derivatives along the curve. This is the canonical L-curve criterion (replacing the earlier box-rescale heuristic); on this kernel it smooths more heavily than GCV, so it is offered as an alternative rather than the default.

Provencher F-test (“probability to reject”; the original criterion of [14], Eqs. (3.23)–(3.24); [15] restate it accessibly). Among the sweep of increasingly-smoothed solutions, each is compared to the least-squares reference by the F-statistic

$$F(\alpha) = \frac{V(\alpha) - V_0}{V_0} \frac{N_y - \text{NDF}_0}{\text{NDF}_0}, \quad V(\alpha) = \|\mathbf{M}^{1/2}(\mathbf{A}\mathbf{x}_\alpha - \mathbf{y})\|^2, \quad (24)$$

where V_0 is the **minimum** residual over the sweep — at the reference α_0 , the least-regularized (least-squares) end — N_y the number of data points, and $\text{NDF}_0 = \text{NDF}(\alpha_0)$ the **effective degrees of freedom at that reference**. Provencher’s NDF (his Eqs. 3.15–3.16), $\text{NDF}(\alpha) = \sum_j s_j^2 / (s_j^2 + \alpha^2)$ with s_j the singular values, is exactly the trace of the Tikhonov influence (“hat”) matrix [12] — *not* the raw grid size:

$$\text{NDF}(\alpha) = \text{tr}[\mathbf{A}(\mathbf{A}^\top \mathbf{A} + \alpha^2 \mathbf{L}^\top \mathbf{L})^{-1} \mathbf{A}^\top] = \sum_j \frac{s_j^2}{s_j^2 + \alpha^2}. \quad (25)$$

Crucially NDF_0 is a **single fixed value** (evaluated at α_0), used in both the scaling factor and the F -distribution — not the per- α DOF. For an ill-posed Laplace kernel it is small (Provencher: “surprisingly small even with accurate data”; a handful to a couple dozen for a typical correlogram), so the test’s degrees of freedom are always well defined. $F(\alpha)$ is F -distributed with $(\text{NDF}_0, N_y - \text{NDF}_0)$ degrees of freedom, so its cumulative value $f_c(\alpha) = \text{CDF}_F(F(\alpha); \text{NDF}_0, N_y - \text{NDF}_0)$ is Provencher’s PROB1, the “probability to reject”: ~ 0 where the regularizer barely changes the fit (under-smoothed) and ~ 1 where it changes it a lot (Provencher: only when $\text{PROB1} > \sim 0.9$ are there significant grounds to suspect α is too large). The chosen α is the one whose f_c is closest to a user level (Provencher’s default 0.5); a **higher** level tolerates more fit degradation and selects a **smoother**, more-parsimonious distribution, a lower level a rougher one.

Caveat. The F-test treats the N_y residuals as independent, but a single correlogram’s lag channels are correlated (Schätzel 1990), so N_y overstates the independent information and the level is a **guide, not an exact test**. It is offered for comparison with legacy CONTIN

output; the chosen method is recorded on the result and in the export. No single criterion is universally best, so — following Hansen’s advice to *monitor several strategies* ([12], §7.1) — all three selectors are user-selectable and their sweeps are returned for inspection.

- **Lognormal** is a *single-mode parametric* distribution: it assumes the intensity-weighted size distribution is lognormal in R_h and fits its median and log-width to the same $\mathbf{A} \mathbf{x} = \mathbf{y}$ model. On the log-spaced grid the discrete weight is

$$w_i \propto \exp \left[-\frac{(\ln R_{h,i} - \mu)^2}{2\sigma^2} \right], \quad \mu = \ln(\text{median } R_h) \quad (26)$$

with σ the log-width (polydispersity); the $1/R_h$ of the lognormal pdf cancels the log-grid spacing, so the discrete weight carries no extra Jacobian. It cannot resolve multiple modes, but it is robust to noise and always returns a smooth, single-peaked distribution — a good default for a known-unimodal sample.

Caution. Fine structure — small secondary peaks especially — is sensitive to noise and the regularization choice and is not always reproducible. Treat the main peaks as reliable; cross-check width against the cumulant PDI and the parametric fits.

3.3 Combining Measurements

Γ versus q^2 (multiple angles). For pure translational diffusion the decay rate is proportional to q^2 :

$$\bar{\Gamma} = D q^2 \quad (27)$$

A straight line through the origin gives D (and R_h). Three linearity diagnostics test whether the motion is genuinely diffusive: the fit R^2 ; the intercept of an unconstrained fit (should be ≈ 0 — a nonzero value indicates a non-diffusive contribution); and the trend of $D_{\text{app}} = \Gamma/q^2$ with q^2 (should be flat — an upturn signals internal/segmental modes). A combined diffusive / non-diffusive flag summarizes these.

Concentration extrapolation. Extrapolating the apparent diffusion coefficient to zero concentration removes inter-particle interactions:

$$D_{\text{app}}(c) = D_0 (1 + k_D c) \quad (28)$$

The intercept D_0 gives the infinite-dilution $R_h,0$; the slope gives the diffusion interaction parameter k_D (positive in good solvents, negative in poor ones).

Both are run from the **DLS tab** method dropdown (“ Γ vs q^2 (sample, multi-angle)” and “ D vs c (sample, multi-conc.)”): they operate on **all** of the selected sample’s DLS measurements (of the current Mw fraction), not just the one in view.

3.4 Depolarized DLS (DDLS): rotational diffusion

For an **optically anisotropic** scatterer (a rigid rod, or an anisotropic particle), the **depolarized** (VH) correlogram decays faster than the polarized (VV) one because molecular *reorientation* adds to the decay. In the small- qL regime the two field (g_1) decay rates are

$$\Gamma_{VV} = q^2 D_t, \quad \Gamma_{VH} = q^2 D_t + 6D_r, \quad (29)$$

where D_t is the translational and D_r the **rotational** diffusion coefficient [16], [17]. Subtracting isolates the rotation, and its inverse is the reorientation time:

$$D_r = \frac{\Gamma_{VH} - \Gamma_{VV}}{6}, \quad \tau_{\text{rot}} = \frac{1}{6D_r}. \quad (30)$$

Each correlogram is fitted for its Γ with the cumulant engine (§3.1); since a homodyne *intensity* correlogram decays at 2Γ , the field rate Γ is what enters Eqs. (29)–(30) — the factor of 2 is handled by the fitter, not double-counted here. Across angles, D_t is taken from the **VV** channel (Γ_{VV} vs q^2 through the origin — the whole VV signal, cleanly measured), and D_r from the mean of the per-angle differences. The $6D_r$ term is q -independent, so a plot of Γ_{VH} vs q^2 is a line of slope D_t and intercept $6D_r$ — the translational/rotational separation at a glance.

Validity. Eq. (29) is a single exponential only for $qL \lesssim 3$ (above it, higher rotational modes and intramolecular interference appear; [17]). If the rod length L is supplied, qL is evaluated per angle and the result is flagged when it exceeds the limit.

Uncertainty. A single angle (one VV + one VH correlogram) carries **no** $\pm$ (one correlogram per channel; [18]). Across angles the points are independent, so D_t gets the through-origin regression SE and D_r the spread ($SD/\sqrt{n}$) of the per-angle estimates — both labeled statistical-only (§6).

Practical note. The VH signal is intrinsically **weak** (the matrix/solvent scatters polarized), so depolarized acquisition needs a clean analyzer and long counting; flexible polymer coils barely depolarize, so DDLS targets rods and anisotropic particles. Mapping D_t , D_r to a rod’s length and diameter (Tirado–García de la Torre) is described in §3.5.

Run from the **DLS tab** $\rightarrow$ **DDLS** sub-tab. Tag each correlogram’s polarization (VV/VH) in the **Data tab** (the *Polarization* row); the sub-tab pairs a VV with a VH at each angle automatically. **Replicates** — several VV (or VH) correlograms at the *same* angle — are averaged (their fitted Γ values are meaned) rather than one being dropped; for weak VH, pre-averaging the correlograms themselves via the sidebar *Average correlation functions* action (then tagging the result) gives better signal-to-noise.

3.5 DDLS shape models: dimensions from D_t and D_r

D_t and D_r are **shape-free observables**; turning them into particle *dimensions* requires assuming a geometry, so the outputs below are **model-dependent** — dimensions of an *assumed* shape, never a direct measurement. Two models are offered, and read together they diagnose the shape.

Sphere (one unknown). The rotational diffusion of a sphere of radius R is the Stokes–Einstein–Debye relation, whose inverse gives R :

$$D_r = \frac{k_B T}{8\pi\eta R^3}, \quad R = \left(\frac{k_B T}{8\pi\eta D_r} \right)^{1/3}. \quad (31)$$

For a *true* sphere this equals the translational (Stokes) radius $R_h = k_B T / (6\pi\eta D_t)$, so the program reports both and their ratio $R(D_r)/R_h$: ≈ 1 means a sphere is plausible; a ratio above 1 is the **rod signature** (rotation reports a smaller effective size than translation), flagging that the sphere model does not apply.

Rigid rod (two unknowns). For a cylinder of length L and diameter d (axial ratio $p = L/d$), the Tirado–García de la Torre [19] relations are

$$D_t = \frac{k_B T}{3\pi\eta L} (\ln p + \nu), \quad D_r = \frac{3k_B T}{\pi\eta L^3} (\ln p + \delta_\perp), \quad (32)$$

$$\nu = 0.312 + \frac{0.565}{p} - \frac{0.100}{p^2}, \quad \delta_\perp = -0.662 + \frac{0.917}{p} - \frac{0.050}{p^2}. \quad (33)$$

With two observables and two unknowns the rod is exactly determined. There is no closed-form inverse, so the program solves it numerically: at a trial p , D_t fixes L (Eq. 32, left); the predicted D_r is monotonic in p , so a one-dimensional root find recovers p , then L and $d = L/p$.

Validity. The end-effect polynomials (Eq. 33) are fitted only for $2 < p < 30$; a recovered p outside that range is flagged as an extrapolation, and if no rod can reproduce both coefficients the rod model is reported as not fitting.

Uncertainty. The sphere radius SE follows from D_r by the delta method ($R \propto D_r^{-1/3} \Rightarrow \text{SE}(R)/R = 1/3 \cdot \text{SE}(D_r)/D_r$). This is a single, smooth, monotonic power, so the delta method is accurate at realistic precision and degrades only for large $\text{SE}(D_r)/D_r$; it is **Monte-Carlo-validated** — sampling D_r and re-inverting matches the delta SE to $\sim 1.5\%$ at a stressing 10 % spread (invariant 8; §6). The rod inverse is nonlinear with no closed form, so its L , d SEs are propagated by **Monte Carlo** directly — sampling D_t , D_r from their SEs and inverting each — which is the sampling spread, with no linearization to validate.

Mapping diffusion to dimensions is the heart of why DDLS is run on rods and anisotropic particles. References: Tirado, López Martínez and García de la Torre [19] (rod); Balog et al. [20] (depolarized DLS of anisotropic nanoparticles, the sphere context).

Run from the same **DLS tab** $\rightarrow$ **DDLS** sub-tab: both models compute automatically after you press **Run** (the sphere needs the solvent viscosity).

4. Static Light Scattering (SLS) Analysis

The SLS module extracts the weight-average molecular weight M_w , the radius of gyration R_g , and the second virial coefficient A_2 . It first converts raw intensities to absolute excess Rayleigh ratios, then constructs the Zimm, Berry, or Debye plots.

4.1 Calibration to an Excess Rayleigh Ratio

Raw intensities are placed on an absolute scale in steps: subtract the detector dark count, subtract the pure-solvent scattering (to isolate the excess from the polymer), correct for the scattering volume via $\sin \theta$, apply the refractive-index correction Eq. (3), and scale by a calibration constant kc :

$$\Delta R_\theta = k_c \sin \theta (I_{\text{sample}} - I_{\text{solvent}}) \left(\frac{n_{\text{solvent}}}{n_{\text{standard}}} \right)^2 \quad (34)$$

There is **one** calibration model, used identically for any instrument. The program computes its own kc from a single calibrant measurement — the intensity of a standard liquid (usually toluene), the angle it was measured at, and that standard’s Rayleigh factor (§1.5, in the matching geometry):

$$k_c = \frac{R_{\text{std}}}{I_{\text{cal,net}} \sin \theta_{\text{cal}}} \quad (35)$$

The constant is shown and can be edited. Any value stored in the instrument file is kept only for reference and **never** enters the calculation (a stored kc goes stale after a re-alignment, and vendor Rayleigh factors are often outdated).

Uncalibrated fallback. If no kc can be computed or supplied, the analysis still runs on an arbitrary scale, flagged uncalibrated. Then M_w and the absolute A_2 are marked unreliable — but R_g is unaffected (it comes from the ratio of slope to intercept, where the scale cancels), and so is the calibration-free product $2A_2M_w$ (§4.4). Those remain valid without calibration.

4.2 Single-Concentration (Apparent) Analysis

At one concentration, a **Debye** plot of $Kc/\Delta R_\theta$ against q^2 is linear; the intercept gives an apparent molecular weight and the slope an apparent radius of gyration. These are **apparent** because the intercept still contains the concentration term,

$$\left. \frac{Kc}{\Delta R_\theta} \right|_{q \rightarrow 0} = \frac{1}{M_w} + 2A_2c, \quad (36)$$

i.e. $1/M_w + 2A_2c$, not $1/M_w$. A single angle yields only an apparent M_w and no R_g . Apparent values are never presented as final.

4.3 Zimm and Berry Double Extrapolation

The full analysis combines several concentrations and angles and extrapolates to both zero angle and zero concentration. The first-order Zimm equation,

$$\frac{Kc}{\Delta R_\theta} = \frac{1}{M_w} \left(1 + \frac{q^2 R_g^2}{3} \right) + 2A_2c, \quad (37)$$

is linear in (q^2, c) , so a single global regression $Kc/\Delta R_\theta = a + b q^2 + d c$ gives all three at once:

$$M_w = \frac{1}{a}, \quad R_g = \sqrt{\frac{3b}{a}}, \quad A_2 = \frac{d}{2}. \quad (38)$$

The **Berry** construction uses the square-root ordinate $\sqrt{Kc/\Delta R_\theta} = a' + b'q^2 + d'c$, giving

$$M_w = \frac{1}{a'^2}, \quad R_g = \sqrt{\frac{6b'}{a'}}, \quad A_2 = d' a'. \quad (39)$$

Berry linearizes the high- qRg regime (roughly Mw above ~ 1 MDa, or where qRg approaches 1) better than Zimm. The program reports which construction it used, and the result is marked **thermodynamic**, not apparent.

Spacing constant. The classic grid spreads each concentration's points along the x-axis by $q^2 + k c$, where k is a purely **aesthetic** spacing constant. Set it in the **Spacing k** box (or press **Suggest** for the automatic value, $q^2_{\max}/c_{\max}$); the chosen k appears in the x-axis label. It only changes how spread-out the curves look — Mw , Rg , and A_2 come from the real (q^2 , c) fit and are unaffected.

4.4 Calibration-Free Second Virial Coefficient

A_2 can be obtained without any absolute calibration, from ratios of excess intensities across concentration at a fixed angle. Because only ratios enter, the optical constant K and the Rayleigh-factor calibration cancel:

$$Y(c) = \frac{I(c_{\text{ref}})/c_{\text{ref}}}{I(c)/c}, \quad \frac{\text{slope}}{\text{intercept}} = 2A_2M_w. \quad (40)$$

This yields the product $2A_2Mw$ directly; if Mw is known, A_2 follows. It is valuable when absolute calibration is uncertain, as in low- dn/dc systems.

Uncertainty. The SE on $2A_2Mw = \text{slope}/\text{intercept}$ is propagated from the HC3 covariance of the line (Eq. 50) through the ratio by the delta method (Eq. 54); the A_2 SE is the linear pass-through $/(2Mw)$. Because this is a **ratio of two fitted coefficients** read off a **short concentration ladder**, it is the one platform SE where HC3's small- n conservatism is pronounced: on a typical 4–6-point ladder the reported $\pm$ **meets or exceeds** the true sampling spread by ~ 20 – 40 % (it never under-reports — the safe direction), and it tightens toward the exact value as more concentrations are added ($\approx$ exact by 12–16 points). This is a property of the single-shot variance *estimate*, not of the propagation — the delta method itself is exact here (a generalized-least-squares fit given the true per-point variances recovers the sampling spread to < 1 %). The remedy for a tighter, unbiased $\pm$ is experimental: more concentration points, or repeat runs (§6.4). See §6.1.

4.5 Guinier Plot

A single-concentration route to Rg using the **log intensity** — the counterpart to the Debye plot of §4.2. The Guinier approximation $\Delta R(q) = \Delta R(0) \cdot \exp(-q^2 Rg^2/3)$ linearizes as

$$\ln[\Delta R(q)] = \ln[\Delta R(0)] - \frac{R_g^2}{3} q^2, \quad (41)$$

so a straight-line fit of $\ln(\Delta R)$ vs q^2 gives $Rg = \sqrt{-3 \text{slope}}$ and an apparent Mw from the intercept ($\Delta R(0) = Kc Mw$). Like Debye's, the Guinier Rg is **scale-independent** (it comes from the slope of a log-intensity plot), so it is reliable even on an **uncalibrated** run; only the intercept-derived Mw needs calibration. The approximation is first-order — exact to fourth order in q for near-spherical particles [21] — so it is used strictly in the Guinier regime $qRg \lesssim 1$ [22]; the program applies a slightly looser practical cutoff ($qRg \lesssim 1.3$, overridable) and flags the largest qRg used. For larger species, Berry or Zimm linearize the high- qRg regime better. Reference: [21] (Guinier's law; the approximation originates with Guinier, 1939); see [1] and [22] for accessible derivations of the Guinier plot.

4.6 Depolarization Correction (Anisotropic Scatterers)

An **optically anisotropic** scatterer — one whose polarizability differs along different molecular axes (rigid rods, many nanoparticles, small anisotropic molecules like toluene; flexible coils are very nearly isotropic) — scrambles the polarization of part of the light it scatters. With the **vertically polarized** incident light used by essentially every modern instrument, this appears as a horizontally polarized (*depolarized*) component. Measuring it both enables the **Cabannes correction** that keeps Mw from being overestimated, and yields the **optical anisotropy** as a result in its own right. This is the **static** side of depolarized light scattering; the **dynamic** side (rotational diffusion Dr from the depolarized correlogram) is described in §3.4.

Convention. Two-letter subscripts are (**incident, analyzer**): IVV = vertical incident + vertical analyzer (the ordinary polarized intensity); IVH = vertical incident + horizontal analyzer (the depolarized intensity). The **depolarization ratio** and its natural-light equivalent are

$$\rho_v = \frac{I_{VH}}{I_{VV}}, \quad \rho_u = \frac{2\rho_v}{1 + \rho_v}, \quad (42)$$

where ρ_u is the ratio for **unpolarized** incident light used by the classical literature (the conversion is exact). Physically $0 \leq \rho_v \leq 3/4$: $\rho_v = 0$ is an isotropic scatterer, $\rho_v = 3/4$ the fully anisotropic limit. A value outside this range signals **stray depolarized light** (polarizer/analyzer leakage, reflections, or an over-subtracted dark count) and is flagged, not used.

Cabannes correction (vertical incidence). The anisotropic part inflates the measured VV Rayleigh ratio, so an uncorrected Mw comes out too high. The **isotropic** excess Rayleigh ratio — the one the Zimm/Debye/Berry analysis (§4.2–4.3) should use — is recovered by

$$\Delta R_{\text{iso}} = \Delta R_{VV} \left(1 - \frac{4}{3} \rho_v\right). \quad (43)$$

For an isotropic scatterer ($\rho_v = 0$) this is the identity, so applying it is always safe. The removed anisotropic fraction is $(4/3)\rho_v$.

Optical anisotropy. In Chu's parametrization the apparent molecular weight and the depolarized Rayleigh ratio carry the mean-square optical anisotropy δ^2 ,

$$M_{\text{app}} = M_w \left(1 + \frac{4}{5} \delta^2\right), \quad \delta^2 = \frac{5\rho_v}{3 - 4\rho_v}, \quad (44)$$

so δ^2 is 0 for an isotropic scatterer and grows without bound as $\rho v \rightarrow 3/4$. Eliminating δ^2 between M_{app} and ρv reproduces Eq. (43) exactly — the two routes agree. (δ^2 here is Chu’s normalization; convert between anisotropy conventions through the depolarization ratio, which is convention-independent, not by equating differently-normalized anisotropies.)

Natural-light form (for older datasets). For **unpolarized** incident light the classical Cabannes factor recovers the isotropic part from the total at 90° ,

$$\Delta R_{\text{iso}} = \Delta R_{\text{total}} \cdot \frac{6 - 7\rho_u}{6 + 6\rho_u}. \quad (45)$$

This is provided for reading natural-light data; modern instruments use Eq. (43).

Uncertainty. A single VV/VH intensity pair carries **no** statistical SE on ρv (consistent with the project rule that one measurement is not an ensemble; §6). When **replicate** intensities are available, the SE on ρv is propagated as a ratio of independent quantities (Eq. 55); δ^2 and the Cabannes factor then follow by the delta method (Eq. 54).

References: Chu [1], §8.4.1.A, Eqs. 8.4.7–8.4.10 (vertical-incidence correction, δ^2); Coumou, Mackor and Hijmans [23] (natural-light Cabannes factor); Sivokhin and Kazantsev [3] (toluene ρv ; $\rho u \leftrightarrow \rho v$).

5. Solvent Property Library

Every scattering calculation needs the solvent’s **refractive index** $n(\lambda, T)$ — it sets the scattering vector $q = (4\pi n/\lambda) \sin(\theta/2)$ (Eq. 1) — and its **dynamic viscosity** $\eta(T)$ — it sets the Stokes–Einstein radius $R_h = kBT/(6\pi\eta D)$ (Eq. 2). The library **proposes** both from a chosen solvent, temperature, and wavelength so they need not be typed by hand. The proposal is a **convenience default only**: it is provenance-tagged, freely overridable, and reaches the analysis solely through the user’s explicit commit (architecture invariant #3). The specific refractive-index increment dn/dc is **never** proposed — it stays hand-entered.

Unified forms. One functional form per property serves every solvent. The refractive index is a three-term Cauchy dispersion with a linear temperature correction to a reference temperature T_{ref} (λ in micrometres):

$$n(\lambda, T) = A + \frac{B}{\lambda^2} + \frac{C}{\lambda^4} + \frac{dn}{dT}(T - T_{\text{ref}}) \quad (46)$$

The viscosity is the Vogel–Fulcher–Tammann (VFT) form (η in Pa·s):

$$\eta(T) = A_\eta \exp\left(\frac{B_\eta}{T - T_0}\right) \quad (47)$$

How the coefficients are obtained. They are fitted **once, offline** (a maintainer script) by evaluating each property’s authoritative primary correlation over the validity box and least-squares-fitting Eqs. 46–47 to it. Refractive-index dispersion comes from dedicated Cauchy/Sellmeier measurements; viscosity from IUPAC/ NIST reference correlations — for those, the fit targets the

paper’s own published **recommended** $\eta(T)$ values at (or, below the normal boiling point, indistinguishably close to) 0.1 MPa, so no equation of state is evaluated at runtime. Each solvent’s sources are listed per row in `physics/solvents.py` and in `docs/5. Citation-Index.md`. The evaluators **raise rather than extrapolate** outside a solvent’s stored wavelength/temperature box.

Confidence tiers. **primary** solvents carry authoritative dispersion and reference-correlation data; **estimate** solvents rest on weaker (e.g. bulk-compilation) data. The tier is a **label only** — the runtime evaluates both identically — that gates which front end offers the solvent and drives display badges.

dn/dT provenance. Where a solvent’s dispersion is measured at a single temperature, the dn/dT slope may be taken from a bulk compilation; such a value is tagged **bulk** (versus **measured** for a slope from the primary source). The slope enters n only through the small linear term of Eq. 46.

Uncertainty: shown, not propagated. Each property carries a display-only tolerance (an absolute n uncertainty; a relative η uncertainty). It is deliberately **not** propagated into any analysis standard error. Per the uncertainty discipline (§6.5, invariant #8) the reported $\pm$ is the statistical/regression uncertainty and explicitly **excludes** calibration-class systematics — and a proposed n/η is exactly that class of systematic. The tolerance informs the user’s judgement; it never travels inside a fitted $\pm$.

The tolerance is per-condition. The library evaluates an honest uncertainty *at the requested wavelength and temperature*, not one flat figure for the whole box. For the refractive index the absolute uncertainty is the quadrature

$$\sigma_n(\lambda, T) = \sqrt{u_{\text{src}}^2 + r_\lambda(\lambda)^2 + r_T(T)^2 + (u_{\text{slope}}(T - T_{\text{ref}}))^2} \quad (48)$$

and for the viscosity the relative uncertainty is

$$\sigma_{\eta, \text{rel}}(T) = \sqrt{u_{\text{src, rel}}^2 + r_{T, \eta}(T)^2} \quad (49)$$

Every component is a real, sourced quantity — nothing about the shape is invented:

- $u_{\text{src}} / u_{\text{src, rel}}$ — the **source’s stated accuracy** (the dispersion measurement’s absolute accuracy; the viscosity correlation’s stated relative uncertainty over the served range). Where a deliberate extra margin was adopted at sourcing time (e.g. glycerol’s hygroscopy allowance), it stays inside this flat floor.
- $r_\lambda(\lambda)$, $r_T(T)$, $r_{T, \eta}(T)$ — piecewise-linear **upper envelopes of the unified-form fit residual** [Eq. 46/47 – source] along each axis, evaluated where the source supplies truth and clamped flat beyond it.
- $u_{\text{slope}}(T - T_{\text{ref}})$ — standard propagation of the dn/dT **slope uncertainty** (a documented per-source allowance: ~5 % for cross-validated slopes, ~8 % for single-source measured, and 15 % for bulk values; **water is 35 %** because its dn/dT is genuinely temperature-dependent rather than uncertain — the constant slope fitted near 20 °C steepens ~67 % by 45 °C per the IAPWS formulation (Harvey *et al.* [24]), and the allowance is sized so the slope term bounds that drift across the whole box), distinct from the model-form residual r_T , so the quadrature is conservative rather than double-counting.

A **bulk/handbook viscosity source** states one figure and no temperature shape, so its band stays honestly **flat** at the record’s floor — a documented non-shape, not an invented taper. The **box-wide scalars** shown in tooltips are the *maxima* of Eqs. 46–47 over the validity box (rounded up by at most one display step, 10^{-4} for n and 10^{-3} for η), so the per-condition band never exceeds them: the band narrows toward each source’s reference conditions only by shedding box-edge pessimism, never below the stated source floor. Three viscosity boxes were clipped to the exact range their source states its uncertainty for (toluene 260–370 K, benzene 288–340 K, methanol 273–343 K) — no band is served past where confidence ends.

Per-solvent uncertainty provenance (values used by Eqs. 46–47; the η floor column shows the *stated* source figure — a record whose sourcing kept a wider deliberate margin carries it on top, and “flat” marks bulk/no-stated-figure sources whose band has no shape):

Solvent	$usrc(n)$	dn/dT slope allow.	η floor $usrc,rel$	η band
water	1.5×10^{-4}	35 %	1 %	shaped
toluene	2×10^{-4}	5 %	0.5 % (260–370 K) (+ margin $\rightarrow$ 0.9 %)	shaped
ethanol	3×10^{-4}	5 %	2.3 % (+ margin $\rightarrow$ 3 %)	shaped
glycerol	3×10^{-4}	8 %	3 % (+ hygroscopy margin $\rightarrow$ 6 %)	shaped
benzene	4×10^{-4}	5 %	1.8 % (+ margin $\rightarrow$ 2 %)	shaped
cyclohexane	2×10^{-4}	5 %	0.5 % (+ margin $\rightarrow$ 2 %)	shaped
<i>n</i> -hexane	6×10^{-4}	15 %	2 %	shaped
methanol	2×10^{-4}	8 %	2 %	shaped
acetone	3×10^{-4}	5 %	5.5 %	shaped
ethylene glycol	3×10^{-4}	8 %	4.9 %	shaped
THF	6×10^{-4}	15 %	6 %	shaped
CCl_4	2×10^{-4}	5 %	— (bulk)	flat 5 %
DMSO	1.5×10^{-3}	8 %	— (no stated figure)	flat 3 %
DMF	3.7×10^{-3}	15 %	— (no stated figure)	flat 3 %
DCM	1.3×10^{-3}	15 %	— (bulk)	flat 5 %
ethyl acetate	1.0×10^{-3}	15 %	— (bulk)	flat 5 %
chloroform	3×10^{-4}	15 %	— (bulk)	flat 5 %
CS_2	4×10^{-4}	15 %	— (bulk)	flat 5 %
acetonitrile	1.5×10^{-3}	15 %	— (no stated figure)	flat 3 %

Substance caveats live here, not in the program. Error sources the program can neither detect nor control are documented rather than warned about in the UI. The prime example is

glycerol, which is strongly hygroscopic: its viscosity rises steeply as adsorbed water is excluded, so the tabulated η is for dry glycerol and a real, water-equilibrated sample can differ by tens of percent (Ferreira *et al.* [25]). Its refractive index shifts with water uptake as well (Jakubczyk *et al.* [26]).

References — refractive index: Daimon and Masumura [27] (water); Samoc [28] (toluene); Moreels, de Greef and Finsy [29] (ethanol, benzene, cyclohexane, acetone); Jakubczyk *et al.* [26] (glycerol); Kozma *et al.* [30] (*n*-hexane); Moutzouris *et al.* [31] and El-Kashef [32] (methanol, dn/dT). Viscosity: Pátek *et al.* [33] (water); Santos *et al.* [34] (toluene); Sotiriadou *et al.* [35] (ethanol), [36] (acetone); Ferreira *et al.* [25] (glycerol); Avgeri *et al.* [37] (benzene); Tariq *et al.* [38] (cyclohexane); Michailidou *et al.* [39] (*n*-hexane); Xiang, Laesecke and Huber [40] (methanol); Mebelli *et al.* [41] (ethylene glycol). Bulk dn/dT : Riddick and Bunger [42]. Estimate-tier additions: Li *et al.* [43] (DMSO, DMF, DCM, ethyl acetate, THF refractive index); Sotiriadou *et al.* [44] (THF viscosity); Samoc [28] (chloroform, CS_2); Moutzouris *et al.* [31] (acetonitrile); Jakubczyk *et al.* [26] (ethylene glycol); Moreels *et al.* [29] (CCl_4); Akmarov *et al.* [45] (measured DMSO dn/dT). Measured (single-lab) viscosities from the pure-component endpoints of binary-mixture studies: Grande *et al.* [46] (DMSO, acetonitrile) and Nikam and Kharat [47] (DMF). Remaining viscosities are bulk (CRC Handbook, Riddick [42]).

6. Uncertainty Estimates

An uncertainty is part of an analysis, not decoration. The program reports the **statistical** (regression) uncertainty of a fit — and **only** where the fitted points are genuinely independent measurements. Where no honest uncertainty exists, **none is reported**. Two scope rules follow from this:

1. **Statistical only.** Every $\pm$ below is the spread implied by the scatter of the fitted points. It **excludes systematic error** — calibration (the constant kc) and dn/dc in particular. Because $Mw \propto 1/(dn/dc)^2$ and scales with kc , those systematics usually dominate the true error of an absolute Mw or A_2 ; the displayed $\pm$ is a **lower bound** on the total uncertainty and is labeled “ $\pm$ statistical; excludes calibration & dn/dc ” wherever a calibration-dependent quantity is shown. R_g , R_h , the size exponent ν , and the calibration-free $2A_2Mw$ are scale-independent, so their statistical $\pm$ is the whole statistical story.
2. **No uncertainty from a single correlogram.** A correlogram’s lag channels are strongly correlated [18], so an ordinary least-squares covariance from **one** correlogram assumes independence it does not have and **under-reports** the truth. ISO 22412 [48] therefore derives DLS dynamic uncertainty from **repeat measurements**. Accordingly the single-correlogram dynamic fits (cumulant, single/double exponential, KWW, lognormal) report **no $\pm$** . The honest dynamic $\pm$ instead comes from **replicate averaging** (§6.4): when you average true repeats, the spread of the per-replicate fits gives an ISO 22412 SD/SEM. The cumulant PDI and the fit RMS remain as width/goodness diagnostics.

6.1 The estimator: HC3 heteroscedasticity-consistent standard errors

For a linear model $\mathbf{y} = \mathbf{X}\mathbf{b}$ with n points, p parameters and residuals e_i , the parameter covariance is the **HC3 “sandwich”** estimator [49], [50]:

$$\text{Cov}(\mathbf{b}) = (\mathbf{X}^\top \mathbf{X})^{-1} \left[\sum_i \frac{e_i^2}{(1 - h_{ii})^2} \mathbf{x}_i \mathbf{x}_i^\top \right] (\mathbf{X}^\top \mathbf{X})^{-1}, \quad h_{ii} = \mathbf{x}_i^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i. \quad (50)$$

Unlike the textbook $s^2(\mathbf{X}^\top \mathbf{X})^{-1}$, HC3 does **not** assume the points share one error variance, so it does not under-report when point precision varies across angles or concentrations (e.g. a per-angle Γ whose precision changes with q). It reduces to the ordinary formula under equal-variance errors and is mildly **conservative** at small n (Monte-Carlo-verified here to meet or exceed the true sampling spread — it never under-reports). When the residual degrees of freedom are too few ($n - p < 2$) no SE is reported, since one residual cannot honestly support a spread (and HC3 blows up as $n - p \rightarrow 1$). The standard error of a coefficient is the square root of its diagonal, e.g. for a straight line $y = a + b x$:

$$\text{SE}(b) = \sqrt{\text{Cov}_{bb}}, \quad \text{SE}(a) = \sqrt{\text{Cov}_{aa}}. \quad (51)$$

That small- n conservatism is largest for a **ratio of two fitted coefficients on a short, high-leverage design** — the calibration-free $2A_2Mw = \text{slope}/\text{intercept}$ (§4.4) is the platform’s clearest case: at a 5-point ladder the endpoints carry leverage ≈ 0.5 – 0.7 , HC3 up-weights their residuals by $(1-h)^{-2}$, and the reported $\pm$ runs ~ 20 – 40 % above the true sampling spread. This is *only* the residual-based variance estimate being cautious — the underlying delta-method propagation is exact (a GLS fit given the true per-point variances matches the sampling spread to < 1 %), and it is not a shared-reference/common-mode effect (fitting the equivalent unshared form $c/\Delta R(c)$ gives an identical SE). The $\pm$ is therefore honest-but-loose in the safe (never-under-reporting) direction; it tightens toward exact as the concentration ladder lengthens, and an unbiased $\pm$ comes from replicates (§6.4). Consistent with invariant 8: where a single-shot SE rests on few, high-leverage points, the achievable honest target is *conservative and disclosed*, with more points / replicates the route to unbiased.

A through-origin slope $y = b x$ (used for $\Gamma = D q^2$) uses the same sandwich with the one-column design $\mathbf{X} = \mathbf{x}$, giving $\text{SE}(b)$ from the scalar covariance.

The Zimm/Berry global fit is multilinear — ordinate $= a + b q^2 + d c$ with design $\mathbf{X} = [1, q^2, c]$ — and uses the same $\text{Cov}(\mathbf{b})$ of Eq. (50) for the full 3×3 covariance:

$$\text{ordinate} = a + b q^2 + d c, \quad \mathbf{X} = [\mathbf{1} \quad \mathbf{q}^2 \quad \mathbf{c}]. \quad (52)$$

6.1.1 Classical OLS standard errors (opt-in, for comparability) Settings offers a switch from HC3 (the default) to the **classical homoscedastic OLS** covariance [51], the one reported by textbooks, classical light-scattering software and spreadsheets:

$$\text{Cov}_{\text{OLS}}(\mathbf{b}) = s^2 (\mathbf{X}^\top \mathbf{X})^{-1}, \quad s^2 = \frac{\sum_i e_i^2}{n - p}. \quad (53)$$

This assumes every point shares one error variance. Choose it **only** to reproduce a result from classical software or a published table *like-for-like* — not as a general improvement. On a short, high-leverage design it **under-reports**: on the 5-point calibration-free A_2 ladder above, OLS returns ≈ 0.90 of the true sampling spread (and modeled-weight WLS ≈ 0.92), where HC3 sits at

≈ 1.3 on the safe side; the OLS deficit shrinks toward 1.0 as points are added. Under-reporting is the one direction the platform never takes by default, which is why **HC3 is the default and OLS is an informed, recorded opt-in** (architecture invariant 8, clause A). The selected estimator is recorded on every result that carries a regression $\pm$ and written into the export — an OLS $\pm$ column's *Comments* cell reads **SE: classical OLS** (HC3 is the silent default). The switch changes only the $\pm$; point estimates (Mw , Rg , D , Rh) are identical either way. Everything downstream (the delta method of §6.2, the ratio and power-law SEs) propagates whichever covariance was selected.

6.2 Propagating a covariance to a derived quantity (delta method)

A scalar function $f(\mathbf{b})$ of the fitted coefficients has, to first order, the standard error given by the **law of propagation of uncertainty** [52]:

$$\text{SE}(f) = \sqrt{\mathbf{J}^\top \text{Cov}(\mathbf{b}) \mathbf{J}}, \quad \mathbf{J} = \nabla_{\mathbf{b}} f. \quad (54)$$

This delivers every SLS SE from Eq. (50)/(52) with the appropriate Jacobian $\mathbf{J}$:

Quantity	Relation	Jacobian $\mathbf{J}$ (order of params)
Mw (Zimm)	$Mw = 1/a$	$[-1/a^2, 0, 0]$
A_2 (Zimm)	$A_2 = d/2$	$[0, 0, 1/2]$
Rg (Zimm)	$Rg = \sqrt{3b/a}$	$[-Rg/(2a), Rg/(2b), 0]$
Mw (Berry)	$Mw = 1/a'^2$	$[-2/a'^3, 0, 0]$
A_2 (Berry)	$A_2 = d'a'$	$[d', 0, a']$
$Mwapp$ (Debye)	$= 1/\text{intercept}$	$[-1/\text{intercept}^2, 0]$
$Rgapp$ (Debye)	$= \sqrt{3 \text{slope}/\text{intercept}} \cdot Mw$	$[-Rg/(2 \cdot \text{intercept}), Rg/(2 \cdot \text{slope})]$
Rg (Guinier)	from slope of $\ln \Delta R$ vs q^2	$\text{SE}(Rg) = (3/2 Rg) \cdot \text{SE}(\text{slope})$
$2A_2 Mw$ (cal-free)	$= \text{slope}/\text{intercept}$	$[-\text{slope}/\text{intercept}^2, 1/\text{intercept}]$

For a **ratio** of two *independent* quantities $r = a/b$ (used for $\rho = Rg/Rh$, with Rg from SLS and Rh from DLS — measured independently), Eq. (54) reduces to the familiar fractional sum in quadrature:

$$\text{SE}(r) = \left| \frac{a}{b} \right| \sqrt{\left(\frac{\text{SE}(a)}{a} \right)^2 + \left(\frac{\text{SE}(b)}{b} \right)^2}. \quad (55)$$

ρ is given a $\pm$ **only when both** Rg and Rh carry an SE (i.e. both came from a regression — a Zimm/Berry Rg with a Γ - q^2 or D - c Rh); a single-cumulant Rh has no SE, so ρ then prints without $\pm$.

For a **power map** $f = k x^p$ (used for $Rh \propto D^{-1}$, so $p = -1$), Eq. (54) gives a fractional relation:

$$\frac{\text{SE}(f)}{|f|} = |p| \frac{\text{SE}(x)}{|x|}, \quad \text{so} \quad \frac{\text{SE}(R_h)}{R_h} = \frac{\text{SE}(D)}{D}. \quad (56)$$

Both the ratio (Eq. 55) and the reciprocal power map (Eq. 56) are **first-order** results, exact only while the *denominator's* fractional error is small. Monte-Carlo checks against the sampling spread confirm they track it to a few percent at realistic precision (a denominator coefficient of variation $\lesssim 0.05$, where a multi-angle Γ - q^2 fit sits) and turn mildly **anti-conservative** — under-reporting by roughly 5–10 % — once the denominator CV reaches ≈ 0.15 , with the heavy tail of $1/(\text{denominator})$ taking over beyond ≈ 0.3 . The direction is set by which quantity is noisy: $\rho = Rg/Rh$ at a viscous-solvent Rh CV of 0.10–0.15 under-reports a few percent, whereas $\rho v = IVH/IVV$ is immune (its noisy channel is the numerator). As everywhere in §6, the honest remedy for a weak denominator is **more points or replicates** — a smaller denominator CV — not a wider algebraic formula. The reciprocal Rh SE (Eq. 56) is carried unchanged wherever Rh follows from D : the Γ - q^2 and D - c extrapolations, and the depolarized Rh_t from D_t .

The cross-sample scaling exponents (ν in $Rg/Rh \propto Mw^\nu$, the slope in $A_2 \propto Mw^{\text{slope}}$) are slopes of a log–log regression, so their SE is Eq. (51) applied to $(\log Mw, \log y)$.

6.3 A consistency check, not an uncertainty: the Zimm two-route Mw

The Zimm grid is extrapolated to the intercept along two independent routes — the $c \rightarrow 0$ line and the $q \rightarrow 0$ line — which must meet at the same $1/Mw$. The program reports both intercept-derived weights and their relative gap:

$$\text{agreement} = \frac{|M_w^{(c \rightarrow 0)} - M_w^{(q \rightarrow 0)}|}{\frac{1}{2}(M_w^{(c \rightarrow 0)} + M_w^{(q \rightarrow 0)})}. \quad (57)$$

This is a **diagnostic of extrapolation quality** (curvature, an over-masked grid), not a statistical uncertainty; a gap above ~ 10 % is flagged for inspection (the interface phrases it as the two routes “differ by X %”). On the Zimm/Berry plot both extrapolation lines are drawn **all the way to the axis** ($q^2 \rightarrow 0$ and $c \rightarrow 0$), meeting at the common intercept ($\approx 1/Mw$, marked), so the double extrapolation is visible end to end.

6.4 Repeat measurements: the one DLS dynamic uncertainty (ISO 22412)

Because a single correlogram cannot supply an honest dynamic $\pm$ (rule 2), the program gets one the way ISO 22412 [48] prescribes — from **repeat measurements**. Select a set of true replicates (the same sample re-measured back-to-back, e.g. a 10-run repeat set) in the navigator, right-click, and choose **Average derived results**. Each replicate is fitted independently with the method you pick, and for a parameter x (e.g. Rh) the n fitted values are summarised as a mean and a standard error of the mean:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}, \quad \text{SEM} = \frac{s}{\sqrt{n}}. \quad (58)$$

The reported value is $\bar{x} \pm \text{SEM}$ (none for $n < 2$, where no spread is defined). Unlike a within-correlogram covariance, the x_i from separate runs **are** independent draws, so Eq. (58) does not under-report; it also captures genuine run-to-run drift (alignment, dust, number fluctuations) that one curve cannot see. The averaged $Rh \pm \text{SEM}$ is written into the sample (so $\rho = Rg/Rh$ can use it), never overwriting a hand-entered Rh . This SE was Monte-Carlo validated: across many

synthetic n -replicate sets the reported $\text{SD}/\sqrt{n}$ matched the actual SD of the set-means to $\approx 1\%$ and the mean estimate was unbiased.

The companion action, **Average correlation functions**, instead takes the channel-by-channel mean of the replicate $g_2(\tau) - 1$ curves (requiring an identical lag grid) to produce a cleaner *point-estimate* correlogram you can fit like any measurement; it carries no $\pm$ of its own — the $\pm$ comes from Eq. (58).

Multi-peak (distribution) methods. *Average derived results* also accepts the distribution methods (NNLS / CONTIN / lognormal), which resolve a *variable* number of size-distribution peaks. These are averaged **by order of appearance**: each replicate’s peaks are sorted by R_h , and position 1 (smallest R_h) is averaged across all runs, position 2 across all, etc., each reported as mean $\pm \text{SD}/\sqrt{n}$ (Eq. 58) with its mean intensity weight and how many of the N runs resolved it. Two cautions are built in: (i) when runs **disagree on the peak count** a warning is shown, because a run that misses a peak shifts the positional alignment (inspect the distribution overlay before trusting the per-peak averages); and (ii) NNLS/CONTIN are **ill-posed** — their peak *positions* depend on the regularization and their fine structure is not always reproducible — so the cross-run $\pm$ is an honest **reproducibility spread**, not a calibrated error, and these peaks are **report-only** (never written to the sample’s R_h). To get an averaged multi-population R_h that *can* feed ρ , average the correlograms (above) and fit that single clean curve with CONTIN in the usual way.

6.5 Where uncertainties are and are not reported

Reported (independent points $\rightarrow$ regression SE valid): SLS Zimm/Berry M_w , R_g , A_2 ; Debye/Guinier apparent M_w , R_g ; calibration-free $2A_2M_w$ (and A_2 if M_w is supplied); Γ -vs- q^2 D and R_h ; D -vs- c D_0 , R_h , 0 , kD ; the scaling exponents; $\rho = R_g/R_h$ when both inputs have an SE; and the **replicate-averaged DLS dynamic** parameters (Eq. 58 — the one single-angle dynamic $\pm$).

Not reported (no honest statistical uncertainty): every single-correlogram dynamic fit (rule 2 above), *unless* repeats are averaged via §6.4; the NNLS/CONTIN distributions (ill-posed inversions — no defensible covariance; the “fine structure is not always reproducible” caveat stands); and a single-angle M_w (one datum, no fit).

6.6 How the $\pm$ is written

The displayed value’s precision is set by its uncertainty (PDG convention): the SE is rounded to one significant figure — two if its leading digit is 1 — and the value is rounded to the same decimal place, e.g. $1.234 \times 10^6 \pm 5.6 \times 10^4 \rightarrow (1.23 \pm 0.06) \times 10^6$, and $40.5 \pm 0.83 \text{ nm} \rightarrow 40.5 \pm 0.8 \text{ nm}$. This governs **every result display**, not just the inline $\pm$ — a value shown in a result table (the Cross-Sample ρ / R_g / R_h table, the DLS Summary R_h table, the SLS results) is rounded to the place its own SE supports, so it never implies more confidence than the $\pm$ justifies (e.g. a $\pm 2 \text{ nm } R_g$ is shown as 42, not 41.7).

Where a result has **no honest $\pm$** (a single-correlogram size, an NNLS/CONTIN distribution peak, a single-angle M_w — §6.5 “Not reported” — and every derived number that carries no $\pm$, including Γ , PDI, R^2 and qR_g), we do **not** fabricate an SE to drive the rounding. Instead the value is shown at a modest **fixed number of significant figures** (default 3), set under *Settings $\rightarrow$ Result Formatting $\rightarrow$ No-uncertainty precision*. This one knob governs **every $\pm$ -less result** uniformly, with no exceptions.

The platform’s sig-fig contract, in three tiers:

1. **Exports carry full precision.** Rounding to the appropriate figures is done by the analyst at the end of a calculation chain, not by the program mid-stream.
2. **A value with an uncertainty** is displayed at the place its $\pm$ supports (the PDG rule above) — it is never shown as more certain than the $\pm$ justifies.
3. **A value with no uncertainty** is displayed at the no-uncertainty knob’s precision.

This is not in tension with the “don’t over-report” principle: for a value that carries no uncertainty, *any* sig-fig choice is inherently arbitrary — 3 is a sensible default, and choosing fewer (which does coarsen R^2/qRg , losing their trailing digits) is a deliberate user decision. The honesty of such a number is carried by its apparent / $\pm$ -omitted qualifier (§6.5); this rule sets only its digit count, and the knob never touches the uncertainty-driven place of a value that *does* carry a $\pm$.

Standing rule. Every analysis — existing or newly added, including the planned depolarized scattering (DPLS) module — must state how it treats uncertainty and, when it reports one, document the formula here with a numbered equation, exactly as the analysis itself is documented.

6.7 Numerical health: the ill-conditioning flag

An SLS fit can be *numerically* degenerate even when it is physically sensible: if the extrapolation design is near-collinear — for a Zimm/Berry global fit the columns of $X = [1, q^2, c]$, for a Debye/Guinier plot the pair $[1, q^2]$, for the calibration-free $2A_2Mw$ the pair $[1, c]$ — the linear algebra that inverts $X^T X$ loses precision, and the derived quantity (a $1/\text{coefficient}$ molecular weight $Mw = 1/a$, or $1/a^2$ for Berry, or $\exp(\text{intercept})/Kc$ for Guinier; or the ratio slope/intercept for $2A_2Mw$) is unreliable regardless of its magnitude. This is distinct from an *uncalibrated* result, whose Mw is on an arbitrary scale, and from mere *statistical* thinness, whose uncertainty is honestly carried by the (inflated) HC3 SE.

The platform flags this with the **condition number of the column-normalized design**, $\kappa = \text{cond}(\hat{X})$, where each column of X is scaled to unit norm first. Normalizing is essential: the raw $\text{cond}(X)$ is dominated by the arbitrary *units* of the columns (q^2 in nm^{-2} , c in g/mL), so it would change if concentration were expressed in mg/mL — the normalized κ reflects genuine collinearity only and is invariant to those unit choices. A well-posed design has $\kappa \approx O(1\text{--}100)$; the flag fires when $\kappa > \kappa_{\text{limit}} = 10^{12}$.

The default 10^{12} is a **numerical-health criterion, not a physics threshold and not a magnitude of Mw** (which would violate the “no system-specific magic numbers” rule — a legitimate polymer can reach tens of millions g/mol): float64 carries ≈ 16 significant digits and loses roughly one per decade of κ , so $\kappa = 10^{12}$ leaves only ≈ 4 — the result is then genuinely numerically unreliable — while sitting far above any healthy design. The limit is a documented, overridable parameter (`cond_limit`).

When the flag fires, the platform does **not** blank the value; it is still shown (the number is finite), with Mw and absolute A_2 marked **unreliable** — in the export Comments cell and as a red flag in the SLS tab — advising *add concentrations or angles, or widen their spread*. Rg (a slope/intercept ratio) is not gated. Two companion guards keep the same discipline at the float boundary: a covariance whose propagated variance cancels to ≤ 0 returns **no SE** (never a spuriously tight $\pm = 0$), and a fit whose Mw overflows to non-finite (a Guinier intercept above ≈ 710 , or a Zimm

intercept underflowing to zero at an extreme intensity scale) is likewise flagged unreliable rather than reported as a finite-looking number.

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